

Missed-Thrust Dispersion for a Bang-Bang Electric Thruster

tax-flow project

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1 Introduction

Electric propulsion systems for deep-space and heliocentric missions deliver their impulse through long, near-continuous thrust arcs rather than short impulsive burns. A fundamental reliability question is: *if the thruster switches off unexpectedly — a missed-thrust event — where*

might the spacecraft end up, and how does the answer depend on how often and how long such outages occur?

This document addresses that question for a **bang-bang thruster**: one that is either fully ON or fully OFF, with no intermediate throttle setting. The thruster's ON/OFF state is modelled as a two-state Markov chain that can switch at every 4-day arc boundary, imposing a minimum dwell of 4 days. Three thruster-reliability scenarios are compared, from a highly reliable unit (outages rare and brief) to an unreliable one (nearly half the arcs are off on average). In addition to the random switching pattern, two sources of *execution error* are present: a small, fixed bias in thrust magnitude and a small, fixed pointing offset, both active whenever the thruster is ON.

The computational method decouples the two uncertainty types. The execution errors are handled analytically — they seed a **Taylor polynomial surrogate** that captures the sensitivity of the spacecraft position to those biases without any extra numerical integration. The random outage schedules are then handled by a Monte Carlo over Markov-chain sequences, but each evaluation costs only a cheap polynomial evaluation rather than a full orbital integration. The result is a sequence of probability density histograms over the orbital plane, from which **highest-density confidence bands** are extracted at 1σ , 2σ , and 3σ levels.

2 Mathematical Background

2.1 Taylor Polynomial Surrogates

The core technique is **Differential Algebra (DA)**: replacing real numbers in the equations of motion with truncated Taylor polynomials in a set of free parameters. Here the free parameters are the two execution errors $(\delta_m, \delta_\theta)$. The equations of motion are formally identical to those for real numbers, but every arithmetic operation — including trigonometric functions, square roots, and divisions — is performed on polynomial objects rather than scalars. The result, after integrating one arc, is a polynomial map

$$\Phi_\Delta(\delta_m, \delta_\theta) = \sum_{|\alpha| \leq P} \mathbf{c}_\alpha \delta_m^{\alpha_1} \delta_\theta^{\alpha_2}$$

that sends any pair of execution errors to the state at the end of that arc. The coefficients $\mathbf{c}_\alpha$ are computed *exactly* (not by finite differences) as a by-product of the integration. At truncation order $P = 6$ and two variables, each state component is described by $\binom{6+2}{2} = 28$ coefficients.

Because δ_m and δ_θ are small — a few percent and a few degrees respectively — the degree-6 polynomial is highly accurate over the entire error box, and the truncation error is negligible.

2.2 Arc-by-Arc Composition

The revolution is divided into N arcs. On each arc the thruster state $f_k \in \{0, 1\}$ is fixed, so the right-hand side of the ODE is smooth within each arc: there is no switching during an integration step. The key operation is to **carry the Taylor polynomial across arc boundaries** without explicit composition.

After integrating arc k , the state is a polynomial in the execution errors. This polynomial is then used directly as the initial condition for arc $k + 1$, with only the constant part updated by the arc's dynamics. The integrator processes the Taylor-valued state exactly as it would a real-valued one — the polynomial algebra is transparent to the time-stepping scheme. After N arcs the state polynomial is the composition of all N per-arc maps:

$$\Phi_{N\Delta} = \Phi_{\Delta}^{(N)} \circ \dots \circ \Phi_{\Delta}^{(2)} \circ \Phi_{\Delta}^{(1)},$$

and this composition is performed automatically, arc by arc, at the cost of N integrations of a Taylor-valued ODE — one per arc, each of duration $\Delta = 4$ days.

2.3 Monte Carlo over Outage Schedules

The execution errors are captured by the polynomial surrogate. The Markov outage schedules are handled differently: each schedule (a sequence of 91 ON/OFF decisions) is drawn independently from the chain, and for that schedule the entire arc-by-arc propagation is run once. At the end of each arc, the polynomial is evaluated at $N_{\text{draw}} = 16$ randomly sampled execution-error pairs $(\delta_m, \delta_\theta)$, adding 16 points to the histogram at that snapshot time. With 8000 schedules the histogram accumulates $8000 \times 16 = 128\,000$ points per snapshot, enough for well-resolved confidence contours.

This two-level approach — polynomial over execution errors, Monte Carlo over schedules — is efficient because the polynomial evaluation is orders of magnitude cheaper than an integration: 28 floating-point operations per state component versus thousands of right-hand-side calls.

2.4 Highest-Density Confidence Bands

From each snapshot histogram the confidence bands are extracted as **highest-density regions (HDR)**: the smallest connected area of the plane enclosing a given probability mass. A 2-D

Gaussian-equivalent labelling is used so the band labels carry familiar meaning:

Band	2-D HDR coverage
1σ	39.35%
2σ	86.47%
3σ	98.89%

The threshold density for each band is found by a sorted cumulative-sum walk on the (lightly Gaussian-smoothed) histogram: sort all pixels by descending density, accumulate mass until the target fraction is reached, and use that pixel's density as the contour level. The result is a contour that need not be elliptical — it traces the actual shape of the probability cloud.

3 Problem Formulation

3.1 Orbit and Nominal Thrust Plan

The spacecraft moves in the **planar heliocentric two-body problem** in canonical units: gravitational parameter $\mu = 1$, reference radius $r_0 = 1$ AU, circular speed $v_0 = 1$, and one orbital period $T = 2\pi$ (approximately 1 year). The initial state is the circular orbit:

$$(x_0, y_0, \dot{x}_0, \dot{y}_0) = (1, 0, 0, 1).$$

The **nominal thrust plan** is continuous full thrust at constant magnitude m_{nom} in the prograde direction (thrust angle $\theta_{\text{nom}} = 0$). For the reference spacecraft (1000 kg, 100 mN thruster):

$$m_{\text{nom}} = \frac{F}{M_{\text{sc}} (\mu_{\odot}/\text{AU}^2)} = \frac{0.1 \text{ N}}{1000 \text{ kg} \times 5.9301 \times 10^{-3} \text{ m/s}^2} \approx 0.0169.$$

3.2 The 4-Day Decision Grid

The revolution is divided into **91 arcs of 4 days each**. The thruster may only change state at an arc boundary, so it holds ON or OFF for *at least* 4 days — the minimum dwell constraint. Each arc k carries a binary thrust fraction

$$f_k \in \{0, 1\}, \quad k = 1, \dots, 91.$$

When $f_k = 0$ the arc is a pure ballistic coast; when $f_k = 1$ the thruster fires at full nominal thrust, modulated by the execution errors.

3.3 Uncertainty Model

Two independent uncertainty sources act simultaneously:

Execution errors (per-trajectory, fixed for the whole revolution):

$$\delta_m \sim \mathcal{U}[-2\%, +2\%], \quad \delta_\theta \sim \mathcal{U}[-5^\circ, +5^\circ].$$

These represent the thruster's calibration uncertainty: a constant fractional error in delivered thrust magnitude and a constant bias in the pointing direction. They are fixed for each trajectory — the same miscalibration acts on every ON arc of that particular trajectory.

Outage sequence (per-schedule, random from arc to arc):

$$f_k \in \{0, 1\},$$

drawn from the Markov chain described below. This represents random thruster trips and recoveries.

3.4 Equations of Motion with Zero Dynamics

The execution errors are appended to the physical state as two extra components with identically zero time derivatives — the same zero-dynamics trick used in the reachability analysis:

$$\mathbf{s} = (\delta_m, \delta_\theta, x, y, v_x, v_y), \quad \dot{\delta}_m = 0, \quad \dot{\delta}_\theta = 0.$$

The full equations of motion on arc k are

$$\dot{x} = v_x, \quad \dot{y} = v_y,$$

$$\dot{v}_x = -\frac{x}{r^3} + f_k m_{\text{nom}}(1 + \delta_m) d_x, \quad \dot{v}_y = -\frac{y}{r^3} + f_k m_{\text{nom}}(1 + \delta_m) d_y,$$

where $r = (x^2 + y^2)^{1/2}$ and the thrust direction is

$$\begin{pmatrix} d_x \\ d_y \end{pmatrix} = R(\theta_{\text{nom}} + \delta_\theta) \hat{\mathbf{v}}, \quad \hat{\mathbf{v}} = \frac{(v_x, v_y)}{\|(v_x, v_y)\|}, \quad R(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}.$$

On an OFF arc ($f_k = 0$) the thrust term vanishes and the motion is pure Keplerian. The execution errors still appear in the state but have no effect until the thruster is next ON.

3.5 Two-State ON/OFF Markov Chain

The sequence of f_k values follows a **sticky two-state Markov chain** on $\{\text{OFF}, \text{ON}\}$, transitioning once per 4-day arc boundary. The transition matrix is

$$P = \begin{pmatrix} 1 - p_{\text{recover}} & p_{\text{recover}} \\ p_{\text{fail}} & 1 - p_{\text{fail}} \end{pmatrix},$$

where $p_{\text{fail}} = P(\text{ON} \rightarrow \text{OFF})$ is the probability of an unexpected trip-off per arc, and $p_{\text{recover}} = P(\text{OFF} \rightarrow \text{ON})$ is the probability of recovery per arc. The chain starts ON ($f_0 = 1$) at $t = 0$.

The stationary ON-duty fraction of this chain is

$$\pi_{\text{ON}} = \frac{p_{\text{recover}}}{p_{\text{fail}} + p_{\text{recover}}}.$$

Three scenarios bracket the range of thruster reliability:

Scenario	p_{fail}	p_{recover}	Stationary ON duty
Reliable	0.02	0.60	$\approx 97\%$
Intermittent	0.08	0.30	$\approx 79\%$
Unreliable	0.20	0.15	$\approx 43\%$

In the reliable case, outages are rare ($p_{\text{fail}} = 0.02$, or about one trip every 50 arcs) and short-lived ($p_{\text{recover}} = 0.60$, so on average the thruster is off for $1/0.60 \approx 1.7$ arcs before recovering). In the unreliable case, outages occur every 5 arcs on average and last for $1/0.15 \approx 6.7$ arcs — the thruster is off nearly as often as it is on.

3.6 Propagation Setup

Parameter	Value
Polynomial order P	6
Expansion variables M	2 (δ_m, δ_θ)
State dimension D	6
Error box half-widths	2% (magnitude), 5° (pointing)
Arc duration	4 days
Number of arcs per revolution	91
Numerical integrator	Verner 8(9) embedded Runge–Kutta
Integration tolerance	10^{-12} (absolute and relative)
Monte Carlo schedules	8000
Draws per schedule per snapshot	16
Histogram grid	130×130 bins

4 Results and Findings

4.1 Dispersion Growth over One Revolution

Figure 1 shows the 3σ confidence boundary at each 4-day snapshot, overlaid on the orbital plane for all three reliability scenarios. The grey circle is the ballistic orbit (no thrust ever applied); the green dashed arc is the nominal trajectory (thruster always ON, no execution errors). Each coloured ellipse is the 3σ boundary at one snapshot, coloured by t/T from blue (early) to red (end of revolution). The Sun is at the origin.

The dispersion ellipses form a string of beads that trail the spacecraft around the orbit. Several features stand out:

- In the **reliable** case the beads are small and tightly strung; with only $\approx 3\%$ of arcs OFF, the spacecraft almost always thrusts and the 3σ boundary remains a compact loop close to the nominal arc.
- In the **intermittent** case the beads widen visibly after the first quarter of the orbit. By the final epoch the 3σ ellipse is roughly three times larger than in the reliable case.
- In the **unreliable** case the beads expand rapidly and, crucially, the *mean position* drifts away from the nominal arc toward the ballistic circle. With $\approx 57\%$ of arcs OFF on average, the spacecraft delivers only about half the intended impulse, so its orbit is intermediate

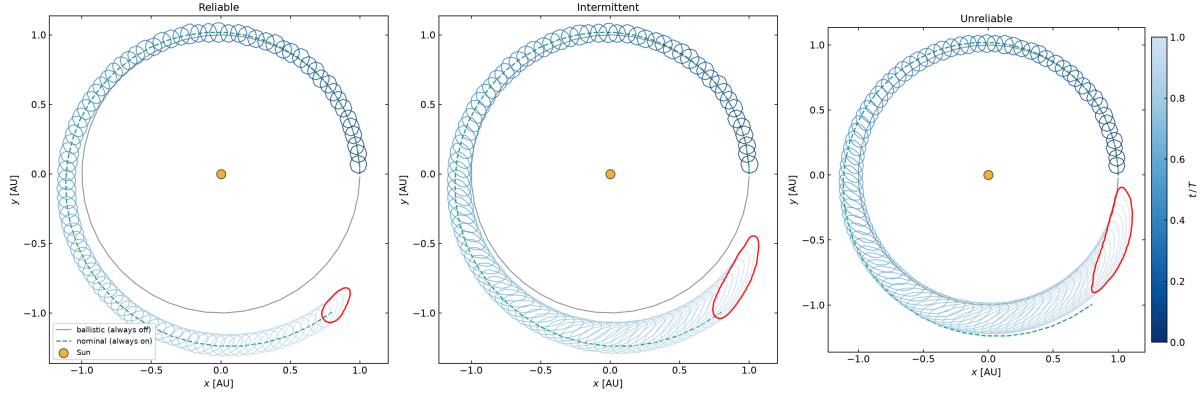


Figure 1: Three-sigma dispersion boundaries at each 4-day snapshot over one heliocentric revolution. Left: reliable thruster ($p_{\text{fail}} = 0.02$, $p_{\text{recover}} = 0.60$). Centre: intermittent (0.08, 0.30). Right: unreliable (0.20, 0.15). Ellipses are coloured by t/T ; the grey circle is the ballistic orbit and the green dashed arc is the nominal (always-on) trajectory.

between the nominal and ballistic references.

4.2 Final-Epoch Nested Confidence Bands

Figure 2 zooms to the end of the revolution ($t = T$), showing the 1σ , 2σ , and 3σ confidence regions together with the nominal endpoint (green star), the ballistic endpoint (grey circle), and the cloud mean (marked $\times$).

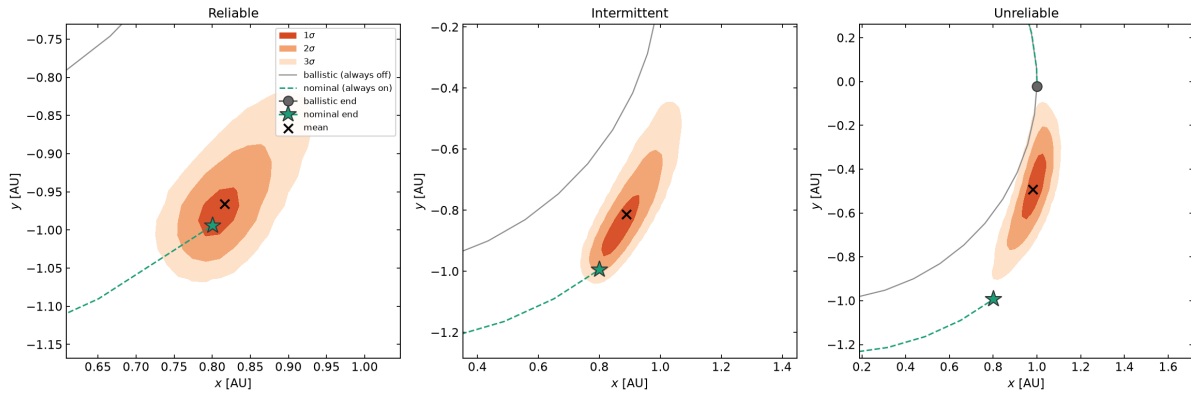


Figure 2: Nested $1\sigma/2\sigma/3\sigma$ confidence regions at the final epoch ($t = T$) for the three reliability scenarios. Green star: nominal endpoint. Grey circle: ballistic endpoint. Cross: sample mean. Note that the plotting scale differs between panels — the unreliable case requires a much wider field of view.

Several points deserve attention:

Shape. The confidence bands are not elliptical. Their elongation is aligned roughly along the orbit arc, reflecting the dominant uncertainty: the spacecraft can be ahead of or behind its expected position depending on *when* missed-thrust events occurred. An early outage (coasting for the first few weeks) puts the spacecraft much further around the orbit than a late outage of the same duration.

Mean displacement. In the reliable case the cloud mean ($\times$) sits very close to the nominal endpoint — the mean number of missed arcs is small. As reliability drops, the mean migrates steadily toward the ballistic endpoint. In the unreliable scenario the mean lies roughly halfway between the nominal and ballistic endpoints, consistent with a $\approx 43\%$ ON duty fraction delivering about half the total impulse.

Scale difference. The three panels are plotted at very different scales: the reliable 3σ band spans ≈ 0.10 AU, the intermittent one ≈ 0.25 AU, and the unreliable one ≈ 0.70 AU. Thruster reliability has a super-linear effect on position uncertainty at the end of the orbit.

Tail toward ballistic. All three distributions show a heavier tail on the side toward the ballistic endpoint (many arcs OFF) than on the side toward the nominal endpoint (no arcs OFF). The all-ON outcome is unique; the many-OFF outcomes span a continuum of intermediate trajectories, pushing probability mass into the tail.

4.3 Duty Cycle and Dispersion Radius versus Time

Figure 3 shows, for each scenario, two time series over the revolution.

Top row — duty cycle evolution. Each stacked area chart shows what fraction of the 8000 simulated trajectories have the thruster ON (green) or OFF (red) at each arc. In the reliable case the ON fraction is nearly constant at 97% from the second arc onward: the chain relaxes to its stationary distribution almost immediately because recovery is fast ($p_{\text{recover}} = 0.60$). In the unreliable case the transition is visible: the chain starts fully ON but within the first 10–15 arcs (40–60 days) the OFF fraction rises to settle around 57%, the stationary level. This transient reflects the chain’s mixing time, $\tau_{\text{mix}} \approx 1/(p_{\text{fail}} + p_{\text{recover}}) = 1/0.35 \approx 2.9$ arcs.

Bottom row — radial dispersion. The $1/2/3\sigma$ envelopes of the radial distance from the ensemble mean grow monotonically with time. This growth is approximately linear because missed-thrust events create velocity offsets that translate into position offsets growing linearly in time (within each arc). Over the full revolution the 3σ dispersion radius reaches:

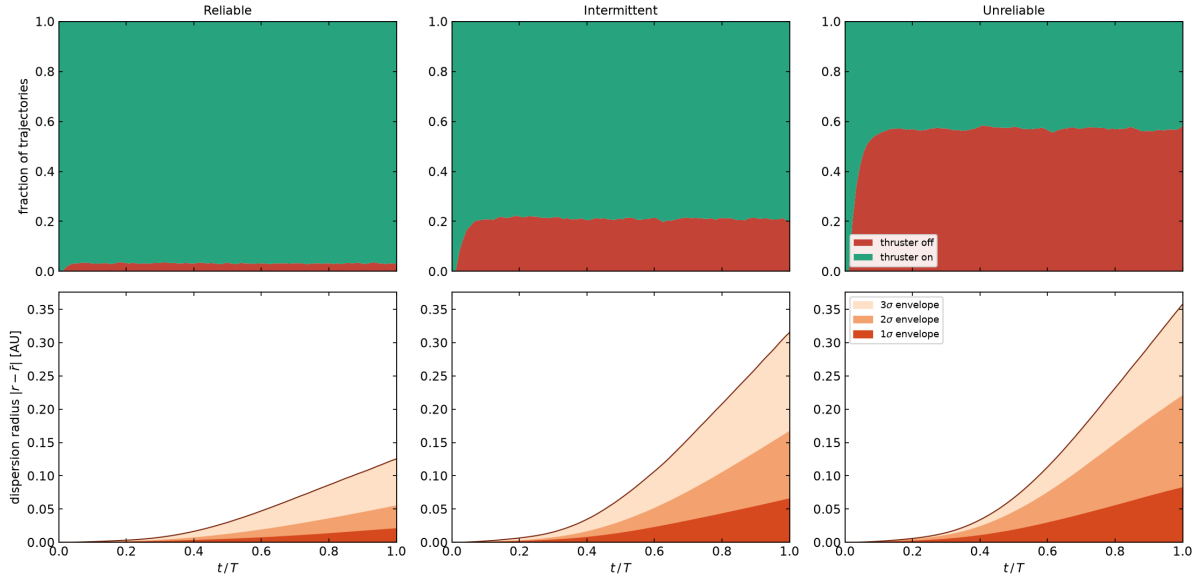


Figure 3: Top row: fraction of Monte Carlo trajectories with the thruster ON (green) versus OFF (red) at each 4-day arc. Bottom row: radial dispersion radius $|\mathbf{r} - \bar{\mathbf{r}}|$ at 1σ , 2σ , 3σ as a function of t/T . Columns correspond to the three reliability scenarios.

Scenario	3σ radial dispersion at $t = T$
Reliable	≈ 0.12 AU
Intermittent	≈ 0.24 AU
Unreliable	≈ 0.36 AU

The ratio is roughly proportional to the mean number of missed arcs, confirming the physical picture: each missed arc contributes an independent velocity impulse offset that accumulates without cancellation over the revolution.

5 Summary

The on/off missed-thrust dispersion problem combines two uncertainty sources with very different mathematical structures:

1. **Execution errors** $(\delta_m, \delta_\theta)$ — small, fixed biases that act continuously on every ON arc. These are handled by a **Taylor polynomial surrogate**: seeding the state as a degree-6 polynomial in the execution errors and propagating it arc by arc, composing the per-arc flow

maps automatically. The polynomial remains accurate across the full error box because the errors are small.

2. **Outage schedules** — binary, random, sequence-valued. These are handled by a **Monte Carlo** over Markov-chain draws. For each schedule the arc-by-arc propagation runs once; the polynomial surrogate is then evaluated cheaply at 16 sample points per arc to fill the histogram without additional integrations.

Three reliability scenarios bracket the design space. The results show:

- **Dispersion is dominated by the outage Markov chain**, not the execution errors. The 3σ band grows roughly in proportion to the mean number of missed arcs; a spacecraft with 57% ON duty has three times the final-epoch uncertainty of one with 97% ON duty.
- **The mean position migrates** from the nominal endpoint toward the ballistic one as reliability drops. For the unreliable thruster the mean shift over one revolution exceeds 0.5 AU — a first-order navigation error, not a statistical tail.
- **The distribution is asymmetric**: heavier tails toward the ballistic endpoint (many outages) than toward the nominal one (no outages), regardless of scenario. Gaussian confidence ellipses would therefore underestimate the risk in the ballistic direction.
- **The DA polynomial surrogate is efficient**: 8000 Monte Carlo schedule evaluations complete in well under an hour, while achieving the accuracy of a direct ODE integration for the execution-error sensitivity.

These properties make the method well-suited for early mission design, where thruster reliability parameters are uncertain and a rapid picture of the position uncertainty at end-of-manoeuve is more informative than a single worst-case trajectory.

6 References

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